$$\tilde{\boldsymbol{q}} = \boldsymbol{q} - \frac{h^2}{12}\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}), \qquad \tilde{\boldsymbol{p}} = \boldsymbol{p} + \frac{h^2}{12}U''(\boldsymbol{q})\boldsymbol{M}^{-1}\boldsymbol{p} \tag{3.9}$$

and inserting these into H yields, after expanding in a Taylor series:

$$\begin{aligned} H(\tilde{\boldsymbol{q}},\tilde{\boldsymbol{p}}) &= T + \frac{h^2}{12}\left(\boldsymbol{p}^T\boldsymbol{M}^{-1}U''\boldsymbol{M}^{-1}\boldsymbol{p} - \nabla U^T\boldsymbol{M}^{-1}\nabla U\right) + \mathcal{O}(h^4) \\ &= \tilde{H}_h(\boldsymbol{q},\boldsymbol{p}) + \mathcal{O}(h^4). \end{aligned}$$

Since $\tilde{H}_h$ is assumed to be constant along the numerical solution, the coordinate transformations have the result of giving an effective order of four for the energy. It turns out that the Takahashi-Imada method is, more generally, an effective 4th order scheme, i.e. for arbitrary quantities, not just the energy [166].

3.4 Perturbative Expansion: Exponentially Small Errors

Let us summarize the general situation with regard to backward error analysis.

Assume a smooth differential equation system

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \boldsymbol{f}(z)$$

with flow map $\mathscr{F}_t$, and a one-step method $\mathscr{G}_h$. We obtain, typically by matching of terms from Taylor expansion, a "modified differential equation" as a series expansion

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \tilde{\boldsymbol{f}}_h(z) = \boldsymbol{f}(z) + h^r\boldsymbol{f}_r(z) + h^{r+1}\boldsymbol{f}_{r+1}(z) + \ldots,$$

where r is the classical order of accuracy of the method. In fact, it is straightforward to show that if numerical method satisfies

$$\mathscr{G}_h(z) - \mathscr{F}_h(z) = h^{r+1}\Gamma_{r+1}(z) + \mathcal{O}(h^{r+2}),$$

i.e. $h^{r+1}\Gamma_{r+1}$ is the leading term in the local error expansion, then we have

$$\boldsymbol{f}_r(z) = \Gamma_{r+1}(z).$$

Now in the case of a Hamiltonian system (with Hamiltonian H) solved using a symplectic numerical method, the above construction can generally be carried out to any desired order by matching of terms in Taylor expansions, assuming sufficient differentiability. In this case it will turn out that

$$\tilde{\boldsymbol{f}}_h = \boldsymbol{J}\nabla\tilde{H}_h,$$